SPACES LOCALLY OF FINITE TYPE OVER A JACOBSON SPACE ARE JACOBSON

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5472-5481, immediately following the statement at 5466-5470.

Editorial note. One cross-chapter pointer (Morphisms) is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Jacobson property for schemes is preserved under locally finite type morphisms. Morphisms, Lemma morphisms-lemma-Jacobson-universally-Jacobson, cited at proof lines 5476-5478. The imported scheme-side result: V -> U locally of finite type with U Jacobson gives V Jacobson.
- Jacobson property defined by an etale cover. definition of Jacobson algebraic spaces invoked at proof lines 5473-5475 and 5479-5481 (Jacobson iff some/every surjective etale scheme cover is Jacobson). Lets the conclusion be read off the single etale cover U.

Source reference. Original reference 02J5: focused-seeds/geometry/S9-decent-jacobson-universally-jacobson/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a Jacobson algebraic space over S. Any algebraic space locally of finite type over X is Jacobson.*

Proof. Let $U \to X$ be a surjective étale morphism where U is a scheme. Then U is Jacobson (by definition) and for a morphism of schemes $V \to U$ which is locally of finite type we see that V is Jacobson by the corresponding result for schemes (Morphisms, Lemma 02J5). Thus if $Y \to X$ is a morphism of algebraic spaces which is locally of finite type, then setting $V = U \times_X Y$ we see that Y is Jacobson by definition. □

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